Early Alignment

$$\frac{d\bar{w}_i^t}{dt} \approx D(w_i^t, \mathbf{0}) - \langle D(w_i^t, \mathbf{0}), \bar{w}_i^t \rangle \bar{w}_i^t$$

Projected gradient ascent/descent on the unit ball for the **continuous** function

$$G(w) = \langle D(w, \mathbf{0}), w \rangle$$

Idea: the neurons weights w_i^t align towards a few directions, given by the critical points of G

18

$h_{\theta^t} \approx 0$ during this phase

- This phenomenon has been used to describe training dynamics for specific examples of data
- Qualitatively described by Maennel et. al (2018)
- No general, rigorous, quantitative description of the phenomenon → our goal

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Main result: early alignment

Assumption: G has no saddle points

Theorem

For any $\varepsilon < \frac{1}{3}$, if $\lambda \leq \lambda^*$, then for $\tau = -\frac{\varepsilon}{D_{\max}} \ln(\lambda)$

1. $\forall t \leq \tau, \forall j \in [m], |a_j^0| \lambda^{2\varepsilon} \leq |a_j^t| \leq |a_j^0| \lambda^{-2\varepsilon}$ (neurons norms don't change much)

2. For any $\alpha_0 > 0$ and neuron j satisfying $\langle \bar{w}_j^0, D(w_j^0, \mathbf{0}) \rangle > -\sqrt{1 - \alpha_0^2} \|D(w_j^0, \mathbf{0})\|$,

$D(w_j^0, \mathbf{0})$ is a critical point of G, along which w_i^τ is aligned:

$$\langle \bar{w}_i^\tau, D(w_i^\tau, \mathbf{0}) \rangle = \|D(w_i^\tau, \mathbf{0})\| - \mathcal{O}_{\alpha_0}(\lambda^{\frac{\varepsilon}{D_{\max}}})$$

λ^* **only depends on the data and** ε (not on m)